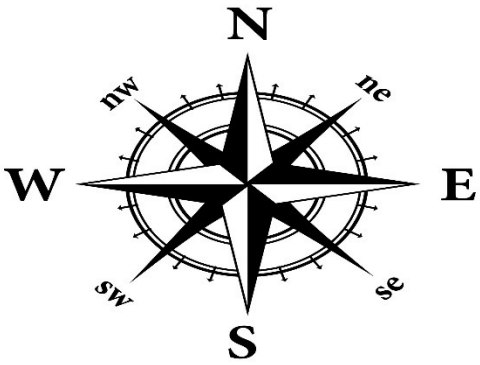
Speed, Velocity and Acceleration

Speed=

Velocity tells us the speed of an object and the direction the object is moving.

So the aeroplane travels at 500Km/h moving west.





Complete this table

Example	Is this velocity? Yes or no?
The car moves North at 100km/h	
The ball rolls at 10m/s	
The boy runs at 12Km/h	
The ship sails South at 10km/h	
The car turns left at 40km/h	
A stone falls towards the ground at 5m/s	

Acceleration is the rate at which speed changes

Or acceleration is the rate at which velocity changes.

A positive acceleration means that something is _____ up.

A negative acceleration means that something is _____ down.

The i_____ speed is the speed the object had before it changed speed.

The f_____ speed is the speed the object had after it changed speed.

Bob's car was travelling at 50km/h (initial speed) and he sped up to 100km/h (final speed)

Complete this table

Situation	Initial speed	Final speed
The aeroplane sped up to a speed of 500km/h from a speed of 200km/h		
The bike was travelling at 20km/h hour and then sped up to 30km/h		
The boy was running at 10km/h. A dog started to chase him so he sped up to 14km/h		

If we know the initial speed and the final speed we can work out the acceleration.

We also need to know how long the change took to happen.

Copy the equation for average acceleration from page 262.

Symbol	What it means
a	Acceleration
v	
u	
t	

Boy rides a bike at 20km/h. He starts to ride faster. After 5seconds he is riding at a speed of 40 km/h. What is his acceleration?

$$a=\frac{v-u}{t}$$

$$a=\frac{40-20}{5}$$

$$a=\frac{20}{5}$$

Acceleration = 4km/h/s

Boy rides a bike at 10km/h. He starts to ride faster. After 10seconds he is riding at a speed of 50 km/h. What is his acceleration?

A cyclist accelerates from 5 km/h to 20km/h in 15 seconds. What is his acceleration?

1. A cyclist accelerates from 0 km/h to 21km/h in 7 seconds. What is his acceleration?

2. A car advertisement states that a certain car can accelerate from zero (0km/h) to 70 km/h in 7 seconds. Find the car's average acceleration.

3. An aeroplane accelerates from 500km/h to 100km/h in 10 seconds. What is its acceleration?

4. You are traveling in a car that is moving at a velocity of 50km/h west. Suddenly, a car 10 meters in front of you slams on its brakes. At that moment, you also slam on your brakes and slow to 5 km/h. Calculate the acceleration if it took 2 seconds to slow your car down.
This is a tricky one. You might get a negative result.

5. A runner covers the last straight stretch of a race in 5 s. During that time, he speeds up from 10km/h to 15km/h. What is the runner's acceleration in this part of the race?